

for this HDD. Once LED1 glows red, switches S2 through S4 are inhibited and therefore you can't activate any HDD.

On the other hand, when the SMPS is 'on' and switch S1 is opened, LED1 glows red. In this state, switches S2 through S4 remain inhibited and no HDD can be connected to the power

supply through them.

Assemble the circuit on a general-purpose PCB and enclose in a box. A large current flows from SCRs and hence PCB tracks should be thick enough. The PCB and connecting wires should be of good quality. Fit the components as shown in the author's prototype (Fig. 3) and mount the PCB

on the spare 13.4cm (5.25-inch) drive slot cover of the CPU cabinet.

EFY note. Once an HDD is activated, you can't deactivate it. To deactivate the HDD, first switch off the power supply to the computer, then wait for a while and switch it on again with the required HDD selector switch closed. ●

AUTOMATIC LIGHT CONTROLLER USING 7806

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Voltage regulator ICs (78xx series) provide a steady output voltage, as against a widely fluctuating input supply, when the common terminal is grounded. Any voltage about zero volt (ground) connected in the common terminal is added to the output voltage. That means the increase in the common terminal voltage is reflected at the output. On the other hand, if the common terminal is disconnected from the ground, the full input voltage is available at the output.

This characteristic is utilised in the present circuit. When the common

terminal is connected to the ground, the regulator output is equivalent to the rated voltage, and as soon as the terminal is disconnected from the ground, the output increases up to the input voltage.

The common terminal is controlled by a transistor, which works as a switch on the terminal. For automatic control of light, a light-dependent resistor (LDR1) is connected to the base of the transistor. In this way, the voltage regulator is able to operate a light bulb automatically as per the ambient light.

To derive the power supply for the circuit, the 50Hz, 230V AC mains is stepped down by transformer X1

to deliver a secondary output of 12V, 250 mA. The secondary output of the transformer is applied to a bridge rectifier comprising diodes D1 through D4, filtered by capacitor C1 and fed to the input terminal of the regulator (IC1).

The common terminal (pin 2) of IC1 is connected to the ground line of the circuit through transistor BC557 (T1). The transistor is biased by R2, R3, VR1 and LDR1. The grounding of IC1 is controlled by transistor T1, while light is sensed by LDR1. Using preset VR1, you can adjust the light-sensing level of transistor T1.

The output of IC1 is fed to the base of transistor T2 (through resistor R4 and zener diode ZD1) and relay RL1. LED1 connected across the positive and ground supply lines acts as a power-'on' indicator.

Normally, the resistance of LDR1 is low during daytime and high during nighttime. During daytime, when light falls on LDR1, npn

